

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

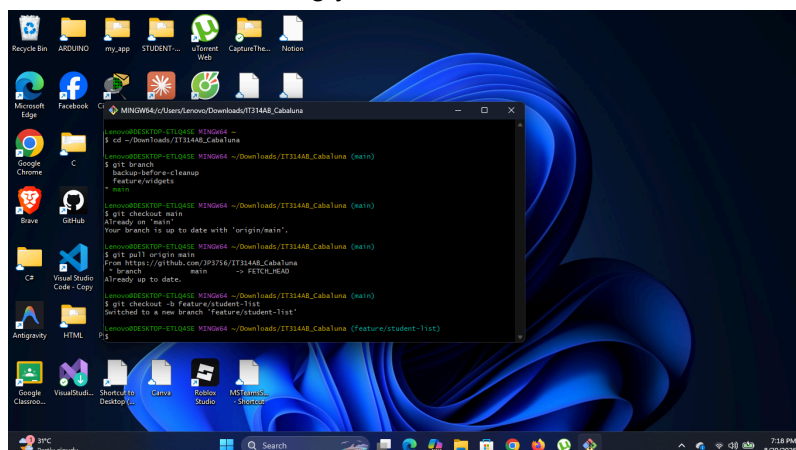
Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch



```
MINGW64 ~/Users/Emorew/Downloads/IT31448_Cabaluna
$ cd ~/Downloads/IT31448_Cabaluna
$ git branch
Backup before cleanup
* feature/widgets
$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
$ git pull origin main
From https://github.com/3P3736/IT31448_Cabaluna
* branch      main       -> FETCH_HEAD
Already up to date.
$ git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'
$
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

The image shows a Windows desktop environment with a Visual Studio Code (VS Code) editor open. The editor's main window displays a JavaScript file named 'student_directory.js'. The code defines an array of student profiles, each represented as an object with properties: 'name', 'course', 'yearlevel', 'age', and 'hobby'. The profiles are for students like Maria, John, Karen, and others, with varying details like course (BSIT, BSBA) and yearlevel (1st, 2nd, 3rd). The VS Code interface includes a sidebar on the left with a file explorer showing a 'student_directory' folder, a search bar at the bottom, and a taskbar at the very bottom with various application icons. The system clock in the bottom right corner indicates the time is 11:03 PM on 8/20/2023.

- Your student list
- At least five student entries

Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

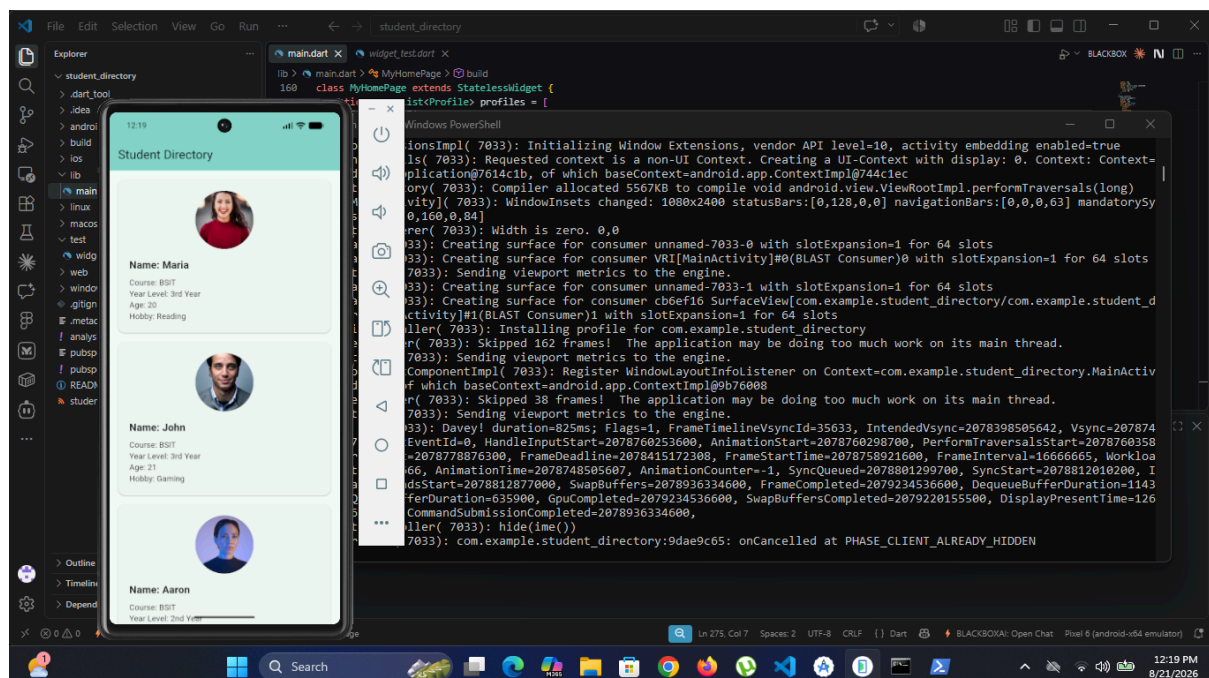
Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:



- Your five manually created student cards (do not change your past code yet)

Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's `ListView`.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

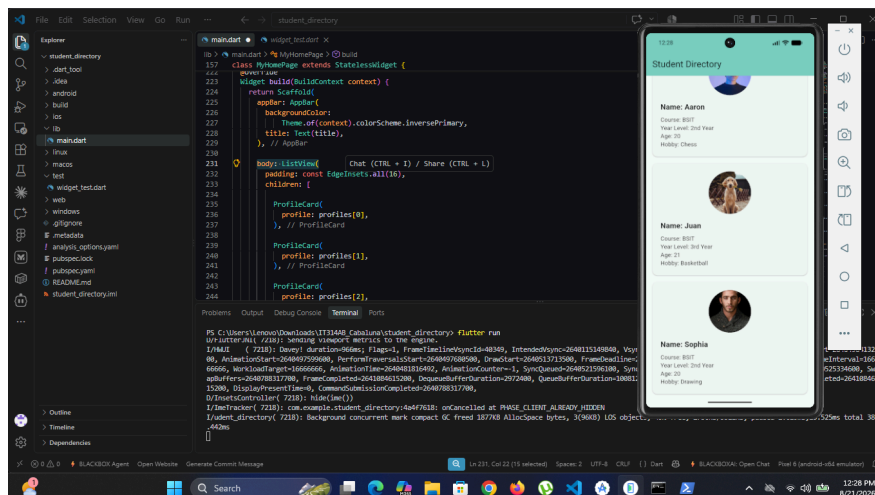
Your application should now allow continuous vertical scrolling.

Test

The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:



- `ListView` in your code
- Top of the application
- Bottom portion showing later students

Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

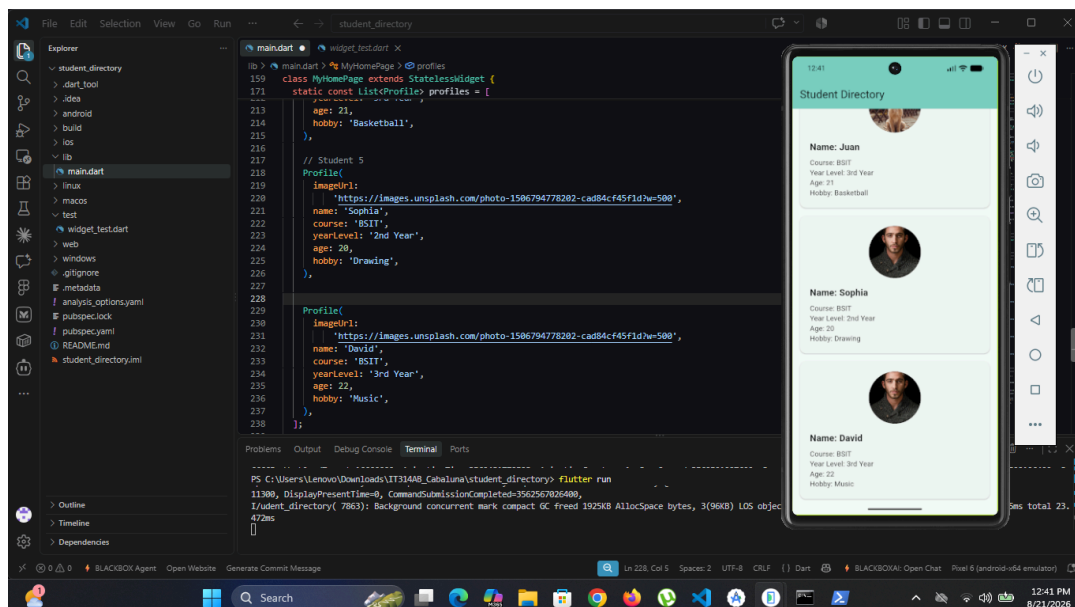
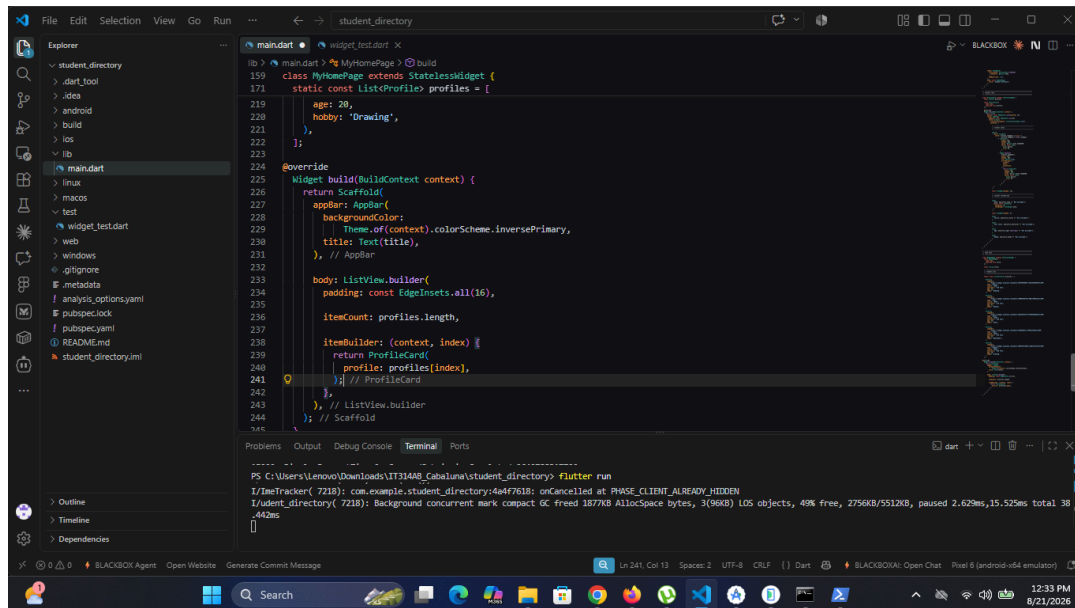
Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

Screenshot:



- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

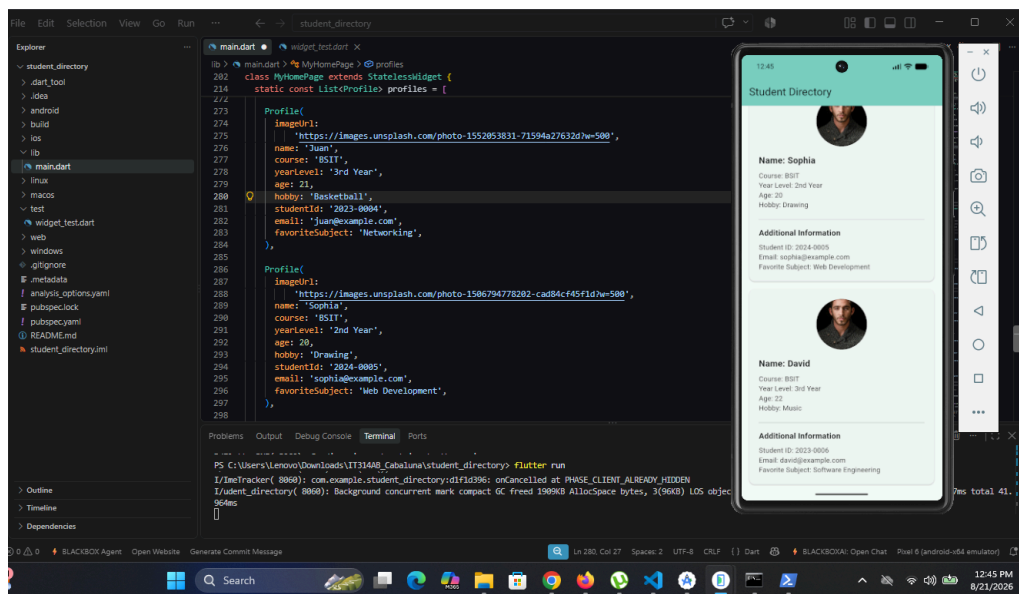
Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:



- Updated student data
- Student Card displaying the additional fields

🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

[]

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

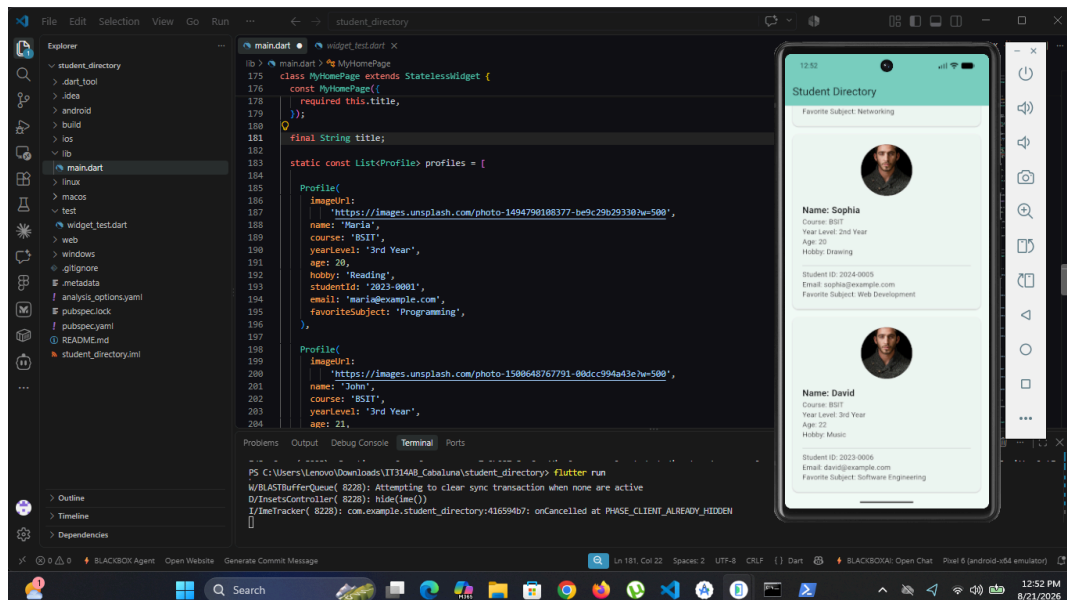
Examples:

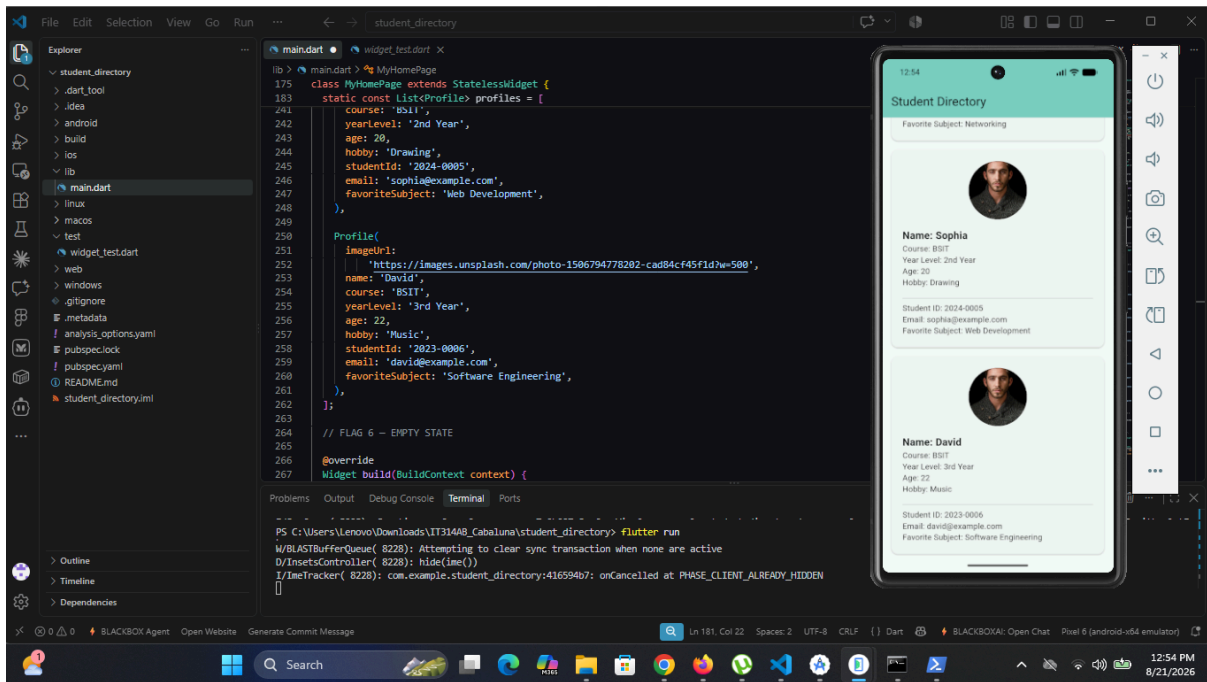
- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:





- Empty student list
- Empty-state UI
- Application still functioning correctly

🚩 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

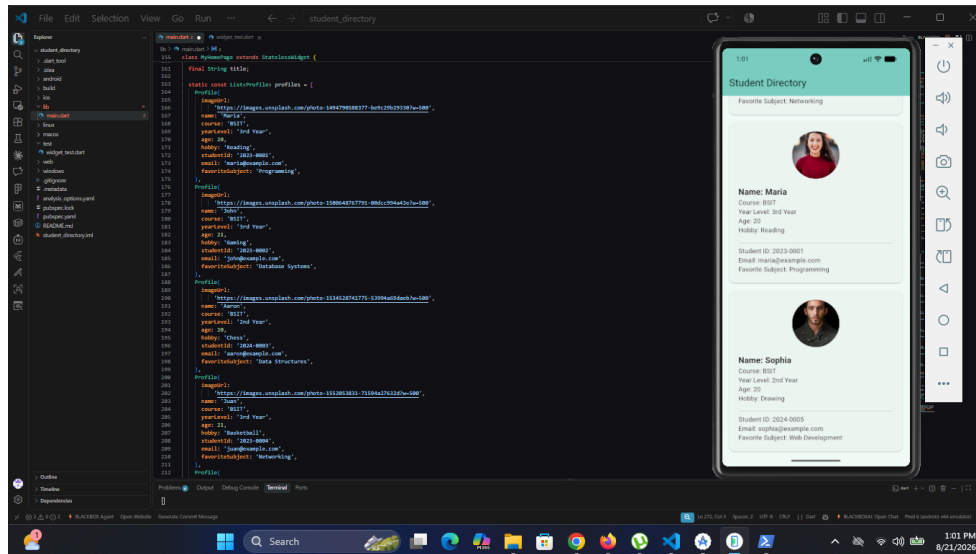
After sorting:

- Aaron
- John

- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:



- Student list before or after sorting
- Application displaying the sorted order

🚩 Flag 8 — Save Your Progress

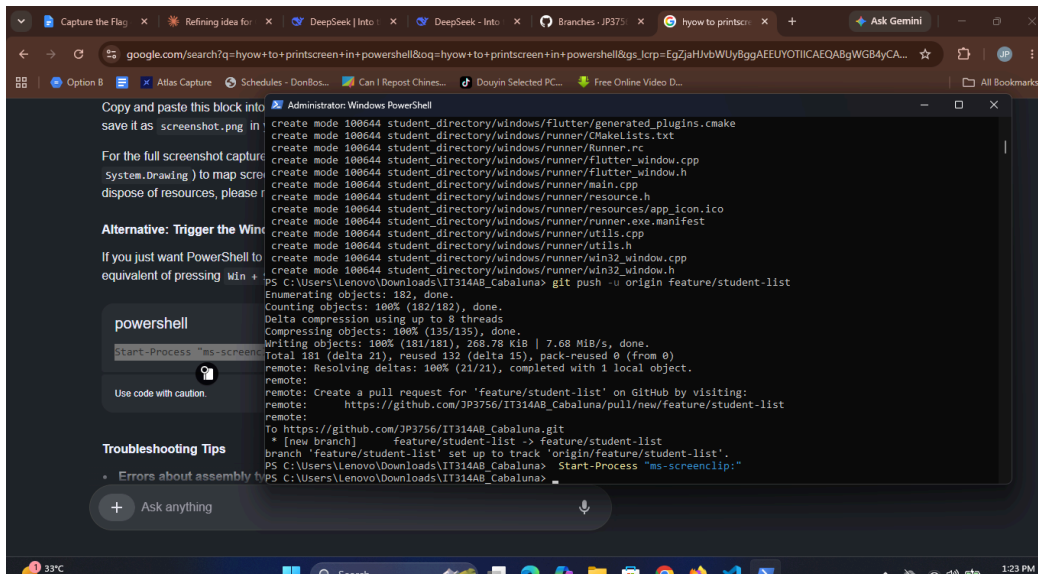
Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:



- Successful commit
- Current branch
- Terminal showing the commit